

BACTIFLOX® -250 / 500 / 750

Antibiotic, antibacterial, gyrase inhibitor

Composition

Bactiflox® -250
Each Lactab contains:
Ciprofloxacin HCl 291.5mg equivalent to ciprofloxacin 250mg

Bactiflox® -500
Each Lactab contains:
Ciprofloxacin HCl 583mg equivalent to ciprofloxacin 500mg

Bactiflox® -750
Each Lactab contains:
Ciprofloxacin HCl 873.3mg equivalent to ciprofloxacin 750mg

Properties/effects

Bactiflox (ciprofloxacin) is an antibiotic from the quinolone group. It has an antibacterial effect against a range of gram-negative and gram-positive bacteria. Bactiflox prevents the transcription of information from the chromosome necessary for normal bacterial metabolism. This leads to a rapid reduction of the reproduction capacity of the bacteria. Bactiflox has a bactericidal effect. Bactiflox is further characterised by the fact that — because of its particular mode of action — there is no cross resistance with any other antibiotic outside the quinolone group. In this way, Bactiflox is in part also effective against bacteria that are resistant to, for example, aminoglycosides, penicilins, cephalosporins, tetracyclines and other antibiotics.

Microbiology

The following pathogens are sensitive (MIC₉₀ < 1 µg/ml)
E. coli, *Shigella*, *Salmonella*, *Citrobacter*, *Klebsiella*, *Enterobacter*, *Serratia*, *Hafnia*, *Edwardsiella*, *Proteus* (indole pos. and neg.), *Providencia*, *Morganella*, *Yersinia*; *Vibrio*, *Aeromonas*, *Plesiomonas*, *Pasteurella*, *Haemophilus*, *Campylobacter*, *Pseudomonas*, *Legionella*, *Neisseria*, *Branhamella*, *Acinetobacter*, *Brucella*; *Listeria*, *Staphylococcus*, *Corynebacterium*, *Chlamydia*.

The following show variable sensitivity (intermediate) (MIC₉₀ = 1-4 µg/ml)
Gardnerella, *Flavobacterium*, *Alcaligenes*, *Streptococcus agalactiae*, *Enterococcus faecalis*, *Streptococcus pyogenes*, *Streptococcus pneumoniae*, *Streptococci of the viridans group*, *Mycoplasma hominis*, *Mycobacterium tuberculosis* and *Mycobacterium fortuitum*.

The following are usually resistant (MIC₉₀ > 4 µg/ml)
Enterococcus faecium, *Ureaplasma urealyticum*, *Nocardia asteroides*.
With a few exceptions, anaerobes are moderately sensitive (e.g. *Peptococcus*, *Peptostreptococcus*) or resistant (e.g. *Bacteroides*). Ciprofloxacin is not effective against *Treponema pallidum*.

Resistance

The development of resistance against ciprofloxacin - as against other quinolones - has been observed in *Staphylococcus spp.* This is particularly true for methicillin-resistant strains of *S. aureus*. Increased resistance has also been described for *Pseudomonas aeruginosa*.

A careful study of literature shows that patients are at particular risk if they require long-term antibiotic therapy as in the case of, e.g. cystic fibrosis or osteomyelitis. The situation is considered similar for patients in particular danger of infection who require intensive antibiotic therapy for prophylactic or therapeutic reasons (e.g. leukaemia patients whose intestinal flora is being selectively suppressed; poly-traumatised or surgical patients who need long-term intensive medical measures). The proportion of resistant populations is subject to large local variations. Regular monitoring of the resistance situation is therefore recommended. There have been reports of fluoroquinolone-resistant *Campylobacter* strains in the context of widespread use in the poultry industry. The mechanism of the development of resistance towards ciprofloxacin has not yet been definitively explained. In certain organisms, however, mutations clearly occur that cause a change of the A subunit of the (ATP-hydrolysing) DNA topoisomerase (also known as DNA gyrase). Resistance can also develop through a change in the outer membrane proteins (porins) and/or other factors that influence the permeability of the organism for the substance.

According to current knowledge, resistance to ciprofloxacin is exclusively chromosomal and therefore cannot be transmitted. There is usually cross-resistance between fluoroquinolones (e.g. ciprofloxacin, enoxacin, norfloxacin and ofloxacin). There is never any cross-resistance between ciprofloxacin and other antimicrobial substances (aminoglycosides, β-lactam antibiotics, sulfonamides including cotrimoxazole, macrolides and tetracyclines).

The above-mentioned impaired permeability may be responsible for the fact that strains of *Pseudomonas aeruginosa* or from the *Enterobacteriaceae* group have rarely been observed to be resistant either towards ciprofloxacin or unrelated substances (e.g. β-lactams and aminoglycosides). Combination of ciprofloxacin with other substances having an antibacterial effect leads mainly to additive or neutral effects in vitro. Ciprofloxacin MIC₅₀ and MIC₉₀ values for a selection of gram-positive and gram-negative bacteria.

MIC₅₀ = minimum inhibitory concentration 50%
MIC₉₀ = minimum inhibitory concentration 90%

In the case of infections caused by moderately sensitive pathogens, a sensitivity test is recommended in order to exclude possible resistance. Sensitivity to Bactiflox can be determined using standardised procedures as, for example, recommended by the National Committee for Clinical Laboratory Standards (NCCLS) with a disk or dilution test. In this case, the NCCLS recommends the following parameters as sensitivity criteria:

	Disk test (5µg) Diameter (mm)	Dilution test MIC (mg/l)
Sensitive	>21	<1
Intermediate	16-20	2
Resistant	<14	>4

Organisms	Number of isolates	MIC50 (mg/l)	MIC90 (mg/l)
Enterobacteria			
<i>Escherichia coli</i>	6315	0.03	0.10
<i>Shigella spp.</i>	1664	0.03	0.04
<i>Salmonella spp.</i>	1770	0.04	0.06
<i>Citmbacterspp.</i>	1602	0.04	0.10
<i>Klebsiella spp</i>	3500	0.04	0.24
<i>Enterobacter spp.</i>	3533	0.04	0.19
<i>Serratiaspp.</i>	1828	0.10	0.70
<i>Hafnia alvei</i>	37	0.02	0.06
<i>Edwardsiella tarda</i>	29	0.06	0.06
<i>Proteus mirabilis</i>	2372	0.03	0.11
<i>Proteus vulgaris</i>	689	0.03	0.05
<i>Providencia alcalifaciens</i>	62	0.02	0.05
<i>Pmvidencia rettgen</i>	310	0.13	1.57
<i>Providencia stuartii</i>	568	0.26	1.69
<i>Morganella morganii</i>	912	0.02	0.12
<i>Yersinia spp.</i>	550	0.03	0.04

Gram-negative pathogens			
<i>Vibrio spp.</i>	162	0.53	0.54
<i>Aeromonas spp.</i>	367	0.02	0.03
<i>Plesiomonas shigelloides</i>	44	0.03	0.03
<i>Pasteurella multocida</i>	20	0.01	0.02
<i>Haemophilus influenzae</i>	1158	0.01	0.02
<i>Haemophilii ducreyi</i>	325	0.01	0.03
<i>Gardnerellavaginalis</i>	230	1.33	2.68
<i>Eikenella corrodens</i>	44	0.01	0.02
<i>Capnocytophaga spp.</i>	249	0.09	0.18
<i>Pseudomonas aeruginosa</i>	6546	0.20	0.71
- <i>malophilia</i>	590	2.22	6.22
- <i>fluorescens</i>	118	0.19	0.81
- <i>cepacia</i>	271	5.04	9.35
- <i>putida</i>	116	0.19	2.52
- <i>pseudomallei</i>	80	3.20	7.05
- <i>acidovorans</i>	49	0.18	0.37
- <i>putrefaciens</i>	32	0.11	0.76
- <i>stutzeri</i>	47	0.14	0.28
<i>Agrobacterium spp.</i>	20	0.06	0.06
<i>Legionella spp.</i>	128	0.30	0.38
<i>Neisseria gonorrhoeae</i>	1899	0.00	0.01
<i>Neisseria meningitidis</i>	266	0.01	0.01
<i>Branhamella catarrhalis</i>	209	0.05	0.10
<i>Acinetobacter spp.</i>	1862	0.36	1.21
<i>Flavobacterium spp.</i>	88	2.17	3.60
<i>Alcaligenes spp.</i>	68	0.52	2.58
<i>Brucella melitensis</i>	179	0.36	0.72
<i>Bordetella spp.</i>	136	0.46	0.54
<i>Campylobacter spp.</i>	1151	0.16	0.68

Gram-positive pathogens			
<i>Staphylococcus aureus</i>	3580	0.32	0.62
- <i>epidermidis</i>	1304	0.18	0.35
- <i>saprophyticus</i>	230	0.38	0.47
- <i>haemolyticus</i>	91	0.20	0.35
- <i>hominis</i>	61	0.17	0.33
<i>Streptococcus pyogenes</i>	524	0.85	2.22
- <i>agalactiae</i>	976	0.70	1.32
- <i>pneumoniae</i>	974	1.31	1.97
- <i>avium</i>	32	1	1.38
- <i>bovis</i>	83	1.85	3.15
- <i>viridans</i>	350	1.60	3.42
<i>Enterococcus faecalis</i>	2726	0.90	1.69
- <i>faecium</i>	131	1.89	3.86
<i>Bacillus spp.</i>	126	0.22	0.62
<i>Lactobacillus spp.</i>	62	4	16
<i>Listeria monocytogenes</i>	449	0.78	1.33
<i>Corynebacterium spp.</i>	176	0.17	0.88
<i>Corynebacterium JK</i>	102	0.54	0.98
<i>Actinomyces spp.</i>	15	1.40	8
<i>Propionibacterium acnes</i>	64	0.96	2.32

Anaerobes			
<i>Bacteroides spp.</i>	3265	4.25	12
<i>Mobiluncus spp.</i>	21	2.33	2.57
<i>Fusobacterium spp.</i>	144	1.91	11
<i>Veillonella spp.</i>	37	0.18	1.96
<i>Peptococcus spp.</i>	182	1.11	2.58
<i>Peptostreptococcus spp.</i>	448	1.15	4.72
<i>Clostridium spp.</i>	1247	5.15	13
<i>Eubacterium spp.</i>	62	4.29	11
<i>Bifidobacterium spp.</i>	14	4	20

Other pathogens			
<i>Borrelis burgdorferi</i>	10	1	2
<i>Chlamydia trachomatis</i>	67	0.99	1.63
<i>Mycoplasma hominis</i>	139	0.64	1.32
<i>Ureaplasma urealyticum</i>	168	5.09	9.61
<i>Mycobacterium tuberculosis</i>	649	0.73	1.62
- <i>avium</i>	135	29	49
- <i>chelonei</i>	87	5.34	27
- <i>fortuitum</i>	139	0.40	1.11
- <i>intracellulare</i>	54	7.85	9.63
- <i>kansasii</i>	118	4.05	30
- <i>marinum</i>	13	1.73	2.70
- <i>xenopi</i>	62	0.70	2.29
<i>Nocardia spp.</i>	164	2.25	19
<i>Rhodococcus spp.</i>	10	0.12	1

Pharmacokinetics

Absorption

After oral administration, ciprofloxacin is absorbed rapidly and almost completely. Because of the first-pass effect, the oral bioavailability is 70-80%. Maximum concentrations in the blood are reached 60-90 minutes after oral administration. Individual doses of 100-750 mg result in maximum serum concentrations of between 0.555 and 3.7 mg/l. The simultaneous intake of ciprofloxacin and milk or dairy products can lead to a reduction in bioavailability of up to 30-36%.

Distribution

In infection sites, i.e. in body fluids and tissues, ciprofloxacin is present in concentrations that are a multiple of those in the serum. The distribution volume of ciprofloxacin is 2-3 l/kg in the steady state. Since the protein binding of ciprofloxacin is low (20-30%) and the substance is mainly present in the blood plasma in non-ionised form, almost the whole quantity of the administered dose can diffuse freely into the extravascular area. In this way, concentrations in certain body fluids and tissues can markedly exceed the corresponding serum levels. Only small ciprofloxacin concentrations reach the cerebrospinal fluid, the maximum concentration being about 6-10% of that of the serum. After oral administration, ciprofloxacin concentrations in bile are many times higher than those in the serum. Ciprofloxacin concentrations in the prostate tissue and fluid are also higher than in the serum after oral administration.

Metabolism / Elimination

The average serum half-life is about 4 hours. Renal excretion after a single oral dose is about 56%. Non-renal excretion of ciprofloxacin takes place mainly through active transintestinal secretion and also through metabolism. About 10-20% of a single dose (oral) are excreted as metabolites.

The individually recovered quantities of metabolites are listed below:

Substance	Oral administration	
	Urine	Faeces
Ciprofloxacin	44.7	25.0
Desethyleno-ciprofloxacin	1.4	0.5
Sulfociprofloxacin	3.7	5.9
Oxociprofloxacin	6.2	1.1
Total	56.0	32.5

A fourth break-down product (formylciprofloxacin) was only found in a few samples at less than 0.1%. Three of the four ciprofloxacin metabolites show comparable or lower antibacterial activity to nalidixic acid. The metabolite formed in smallest quantity (formylciprofloxacin) is also the most active and its efficacy corresponds largely to that of norfloxacin. More than 90% of renal excretion takes place in the first 24 hours. A comparison of the pharmacokinetic parameters on intravenous administration for two months and three months gave no indication of an accumulation of ciprofloxacin and its metabolites.

Kinetics in special clinical situations

In older patients creatinine clearance should be checked since the elimination half-life may be prolonged. In cases of impaired renal function, the dose must be halved or the dosage interval doubled if the creatinine clearance is less than 20ml/min.

Because of the low rate of ciprofloxacin metabolism, accumulation in patients with impaired liver function is improbable. Pharmacokinetic studies on children/adolescents with cystic fibrosis showed that the plasma concentration profiles for them are comparable to those for adults with the respective recommended doses (oral 2x20 mg/kg, parenteral 10 mg/kg).

Indications/possible uses

Consideration should be given to available official guidance on the appropriate use of antibacterial agents. For BACTIFLOX-500 LACTAB: Uncomplicated and complicated infections caused by ciprofloxacin susceptible pathogens.

*Infections of the respiratory tract:
Ciprofloxacin can be regarded as an advisable treatment for pneumonias caused by *Klebsiella*, *Enterobacter*, *Proteus*, *E. coli*, *Pseudomonas*, *Haemophilus*, *Moraxella catarrhalis*, *Legionella* and *Staphylococcus*.

*Infections of the middle ear (otitis media), of the paranasal sinuses (sinusitis), especially if these are caused by Gram-negative organisms including *Pseudomonas aeruginosa* or by *staphylococci*. Infections of the eyes

*Infections of the kidneys and/or the efferent urinary tract
Infections of the genital organs, including adnexitis, gonorrhea, prostatitis and excluding vaginal infections
Infections of the abdominal cavity (e.g. infections of the gastrointestinal tract or of the biliary tract, peritonitis)
Infections of the skin and soft tissue
Infections of the bones and joints

*Sepsis
Infections or imminent risk of infection (prophylaxis) in patients whose immune system has been weakened (e.g. patients on immunosuppressants or have neutropenia)
* BACTIFLOX-500 LACTAB should be only used:
• When *Pseudomonas* is considered AND the patient is allergic to antipseudomonal penicillins/cephalosporin; or
• For resistant organism with no other alternative antibiotics available.
For BACTIFLOX-500 LACTAB only:
Prophylaxis of invasive infections due to *Neisseria meningitidis*.

Children and adolescents
Ciprofloxacin may be used in children for the second- and third-line treatment of complicated urinary tract infections and pyelonephritis

due to *Escherichia coli* (age range applied in clinical studies: 1 – 17 years) and for the treatment of broncho-pulmonary infections of cystic fibrosis associated with *Pseudomonas aeruginosa* (age range applied in clinical studies: 5 – 17 years). Treatment should only be initiated after careful benefit/risk evaluation, due to possible adverse events related to joints and/or surrounding tissues. The clinical trials in children were performed in the indications listed above. For other indications clinical experience is limited.

Dosage/Use Usual dosage

Oral	Individual/daily doses for adults
Simple infections of the lower and upper urinary tract	2x125mg
Infections of the urinary tract: depending on severity	2x250 mg to 2x500 mg
Respiratory infections (e.g. bronchitis)	up to 2x500 mg
Other infections (cf. "Indications")	2x500 mg
Severe infections (e.g. osteomyelitis, respiratory tract infections in cases of mucoviscidosis)	2x750 mg

In cases of acute, uncomplicated gonorrhoea in women and men (urethritis) and of uncomplicated cystitis in women, a single dose of 250 mg is sufficient.

In the case of particularly threatening infections, e.g. recurrent infections in patients with mucoviscidosis, pneumonias, infections of the peritoneum (peritonitis), bones and joints caused by *Pseudomonas*, *Staphylococcus* or *Streptococcus pneumoniae*, the daily dosage must be increased to 2x750 mg if treatment is not parenteral.

Although its efficacy has been proved in clinical studies, Bactiflox is not the medication of first choice for the treatment of suspected or diagnosed pneumonias caused by *Streptococcus pneumoniae*.

In cases of urinary tract infections with Chlamydia, the daily dose should, if necessary, be increased to 2x750 mg and be taken for at least ten days.

In the case of oral administration, the tablets should be swallowed unchewed with liquid. The tablets can be taken at any time, independently of meals. Taking them on an empty stomach accelerates absorption.

The length of treatment depends on the severity of the illness and the clinical and bacteriological course. Treatment should always continue for at least 3 days after the disappearance of fever or clinical symptoms.

Average length of treatment: 1 day for acute, uncomplicated gonorrhoea in men; up to 7 days for infections of the kidneys, urinary tract and peritoneum; for the whole neutropenic phase in patients with weakened defenses; 7 to 14 days for all other infections and at most 2 months for osteomyelitis.

In the case of *Streptococcus* infections, treatment should last at least ten days because of the risk of delayed complications.

Special dosage guidelines

Elderly: Elderly patients should receive a dose as low as possible depending on the severity of their illness and the creatinine clearance.

Children: contraindicated

Renal and hepatic impairment

- Dosage when renal function is impaired
 - With a creatinine clearance of between 31 and 60 ml/min/1.73 m² or a serum creatinine level between 1.4 and 1.9 mg/100 ml, the maximum daily dose for oral administration should be 1000 mg.
 - With a creatinine clearance of 30ml/min/1.73 m² and below and a serum creatinine level of 2.0 mg/100 ml and above, the maximum daily dose for oral administration should be 500 mg.
- Dosage for patients with impaired renal function undergoing haemodialysis Dosage as for 1.2; after the dialysis on dialysis days.
- Impaired renal function + CAPD
Administration of ciprofloxacin film coated tablets as 1x500mg film coated tablet or 2x250mg film coated tablets
- Dosage when liver function is impaired
Ciprofloxacin elimination is only slightly changed when liver function is impaired. With the usual dosage, no dose adjustment is necessary.
- When both liver and kidney functions are impaired, the dosage must be adjusted to the degree of renal function impairment (see 1.1 and 1.2).

Restrictions on use

Contraindications

Bactiflox must not be administered in cases of hypersensitivity to ciprofloxacin or other chemotherapeutic agents of the quinolone type.

Until further results are available, Bactiflox must not be administered to pregnant or lactating women.

Children should not be treated with Bactiflox, nor should adolescents, until they have finished growing, since results from animal experiments have not been able to preclude damage to articular cartilage when growth is still in progress. Concerning mucoviscidosis, see "Precautions".

Warnings and Precautions

The use of ciprofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Adverse Effects/Undesirable Effects). Treatment of these patients with ciprofloxacin should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Prolonged, disabling and potentially irreversible serious adverse drug reactions

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Ciprofloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with ciprofloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.

*[For systemically administered levofloxacin-containing products, the listing of risk factors should additionally include: "in patients receiving daily doses of 1000 mg levofloxacin".]

Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with ciprofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section Adverse Effects/Undesirable Effects).

Ciprofloxacin, like other gyrase inhibitors, is known to cause damage to weight-bearing joints in juvenile animals. Analysis of the safety data for 18 year old patients mainly with cystic fibrosis (mucoviscidosis) showed no evidence of joint/cartilage damage.

In rare cases, ciprofloxacin may cause photosensitivity reactions. Such patients should avoid prolonged exposure to sunlight during ciprofloxacin treatment. If this is not possible, a sun cream with an adequate sun protection factor should be used, and clothing covering the arms and legs, and a hat to shield the face, should be worn.

In epileptics and patients with central nervous system damage (e.g. lowered convulsion threshold, history of convulsions, impaired cerebral circulation, cerebral structural changes or stroke), Bactiflox should be administered only after careful risk-benefit assessment, since these patients are at risk of possible central nervous side effects.

Long-term and repeated administration can lead to superinfections with resistant bacteria or yeast-like fungi.

The possibility of cross-resistance between ciprofloxacin and other fluoroquinolones must be monitored.

Because of possible phototoxic reactions, patients should be advised to avoid excessive exposure to sun.

Since ciprofloxacin is mainly excreted in the urine and to a lesser degree via the hepatobiliary system, care must be taken in the case of patients with impaired renal function.

Crystalluria has been reported (rarely) and patients must therefore be told to drink sufficient quantities of fluid.

Individual reactions impairing the ability to take active part in road traffic or to operate machinery are possible. This effect can be stronger if combined with alcohol intake.

Exacerbation of myasthenia gravis

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

Pregnancy/Lactation

Pregnancy category C

Animal experiments have given no indications of teratogenic effects (malformations), but the possibility of damage to growing cartilage exists.

Ciprofloxacin enters umbilical cord blood and the amniotic fluid. Ciprofloxacin has been found in breast milk in similar concentrations to those in the serum.

For this reason, Bactiflox should not be administered to pregnant or lactating women until further information is available.

Adverse Effects/Undesirable Effects

Side effects as a whole are rare during treatment with ciprofloxacin. The following undesired effects of the medication have been observed:

Gastrointestinal symptoms

2-10% of patients suffered from nausea, diarrhoea, vomiting, digestive disorders, abdominal pain, bloating and loss of appetite. Severe and lasting diarrhoea during or after therapy could be a sign of a serious intestinal disorder (pseudomembranous colitis) that requires immediate treatment. In such cases, Bactiflox must be discontinued at once and suitable diagnostic and therapeutic measures initiated at once (e.g. vancomycin oral 4x250 mg daily). Peristalsis-inhibiting preparations are contraindicated.

Disturbances of the central nervous system

Occasional: dizziness, headaches, fatigue, insomnia, agitation, tremor.

Very rare: peripheral perception disturbances, sweating, uncertain gait, convulsions, rise in intracranial pressure, anxiety states, nightmares, confusion, depression, hallucinations; in isolated cases: psychotic reactions, disturbances in taste and smell, visual disturbances (e.g. diplopia, colour vision). These reactions have occasionally occurred after the first administration. In such cases, Bactiflox must be discontinued immediately.

Hypersensitivity reactions

Occasional: skin reactions such as rashes.

Very rare: itching, medication fever, urticaria, hyperpigmentation, anaphylactic/anaphylactoid reactions (e.g. facial, vascular and larynx oedemas; shortness of breath possibly developing into life-threatening shock), occasionally even after the first administration. In such cases, Bactiflox must be discontinued immediately, and medical treatment (e.g. shock therapy) is necessary.

Punctiform skin bleeding (petechiae), blister formation with bleeding (haemorrhagic bullae) and small nodes (papules) with scabs as expression of vascular involvement (vasculitis), erythema nodosum, erythema exudativum multiforme (minor), Stevens-Johnson syndrome, Lyell's syndrome, interstitial nephritis, hepatitis, liver cell necrosis possibly developing into life-threatening liver failure.

Effects on the cardiovascular system

Tachycardia

Very rare: hot flushes, migraine, fainting

Other

Joint complaints, swollen joints

Very rare: general weakness, muscular pain, tendovaginitis, photosensitivity, temporary reduction of renal function and possibly temporary renal failure, tinnitus, deafness, particularly for high-pitch sounds, usually temporary.

In isolated cases, achillotendinitis has been observed during administration of ciprofloxacin. Isolated cases of, in particular, elderly patients who had been previously treated systemically with glucocorticosteroids have been reported with a partial or complete rupture of the Achilles tendon. Ciprofloxacin should therefore be discontinued where there are signs of achillotendinitis (e.g. painful swelling).

Effects on blood and blood components

Eosinophilia, leucocytopenia, granulocytopenia, anaemia, thrombocytopenia

Very rare: leucocytosis, thrombocytosis, haemolytic anaemia, changed prothrombin values

Local reactions:

Very rare: phlebitis

Local reactions have been observed after intravenous administration of ciprofloxacin. These reactions are more common if the duration of the infusion is 30 minutes or less. They may occur as local skin reactions that disappear quickly after the end of the infusion. Further intravenous infusions are not contraindicated unless there are repeated or more severe reactions.

Influence on laboratory/urine sediment

In patients with liver damage, in particular, there may be a temporary rise in transaminases, LDH and alkaline phosphatase; a temporary rise in urea, creatinine and bilirubin in the serum; in isolated cases: hyperglycaemia, crystalluria and haematuria.

Post Marketing Experience

Exacerbation of myasthenia gravis

Musculoskeletal and connective tissue disorders*

Nervous system disorders*

General disorders and administrative site conditions*

Psychiatric disorders*

Eye disorders*

Ear and labyrinth disorders*

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reaction such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Warnings and Precautions)

Interactions

The simultaneous intake of ciprofloxacin (oral) with iron, sucralfate or antacids and strongly buffered preparations (e.g. anti-retroviral agents) containing magnesium, aluminium or calcium reduces the absorption of ciprofloxacin. For this reason, ciprofloxacin should be taken either 1-2 hours before or 4 hours after such preparations.

This restriction does not apply to antacids of the H₂-receptor blocker type. Simultaneous administration of Bactiflox and theophylline can lead to an undesired rise in the serum theophylline concentration to toxic levels. This can lead to side effects caused by theophylline. If simultaneous intake of the two preparations is unavoidable, the theophylline concentration in the serum should be checked and its dosage reduced accordingly.

Animal experiments have shown that the combination of extremely high doses of quinolones (gyrase inhibitors) and certain non-steroidal anti-inflammatory drugs (such as fenbufen, but not acetylsalicylic acid) can trigger convulsions.

In isolated cases, a temporary rise in the serum creatinine

concentration has been observed with simultaneous administration of ciprofloxacin and ciclosporin. For this reason, the serum creatinine levels of these patients require frequent checking (2 x weekly).

Simultaneous administration of Bactiflox and warfarin can increase the effect of the warfarin.

In isolated cases, simultaneous administration of Bactiflox and glibenclamide can enhance the effect of glibenclamide (hypoglycaemia).

Probenecid influences the renal secretion of ciprofloxacin. Simultaneous administration of probenecid (1000 mg) and ciprofloxacin (500 mg) increased the serum concentration of ciprofloxacin by about 50%, while the elimination half-life remained unchanged, which should be taken into account for patients taking both medications.

Metoclopramide accelerates the absorption of ciprofloxacin, which means that maximum plasma concentrations are achieved more rapidly. The bioavailability of ciprofloxacin is not affected.

Overdosage

In the event of acute overdosage, reversible renal toxicity has been observed in a few cases. Besides the usual emergency measures, the recommendations are to monitor renal function and to administer antacids containing magnesium or calcium that reduce the absorption of ciprofloxacin. Haemodialysis or peritoneal dialysis removes only a small proportion of ciprofloxacin (<10%) from the body.

Sufficient liquid intake is necessary to prevent crystalluria.

Presentation

Bactiflox-250 Lactab: White, round biconvex, film-coated tablet
Packing of 10 Lactab in blister pack

Bactiflox-500 Lactab: White, oblong, biconvex, film-coated tablet with breaking score on one side.
Packing of 10 Lactab in blister pack

Bactiflox-750 Lactab: White, biconvex oblong, film-coated tablet scored on one side
Packing of 10 Lactab in blister pack

Other information

Note: Medicines should be kept out of the reach of children.

Storage: Store dry below 30°C.

Shelf life: 5 years

Expiry: The preparation should not be used after the date marked "EXP" on the pack.

Information date

March 2024

Manufactured by:

Sofarimex, Industria Química e Farmaceutica, Lda
Avenida das Industrias
Alto do Colaride, Agualva
2735-213 Cacem
Portugal
for Acino AG, Am Windfeld 35, 83714 Miesbach
Germany

